Fiscal Year:	FY 2025	**Task Last Updated:**	FY 10/18/2024
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Project Title:	Characterizing and Mitigating the Interactive Impacts of Fatigue- and Altered Gravity-Related Stressors on Sensorimotor, Behavioral, and Operational Outcomes		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders (2) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks (3) **Sleep**:Risk of Performance Decrements and Adverse Health Outcomes Resulting from Sleep Loss, Circadian Desynchronization, and Work Overload		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
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No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
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Flight Program:			
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Task Description:	Crews during lunar orbital and surface missions will be exposed to multiple spaceflight stressors simultaneously (such as fatigue and altered gravity), which can adversely impact their sensorimotor and behavioral functions and subsequently result in performance decrements on manual/vehicle control or extravehicular activities. Despite the growing evidence reported in terrestrial studies on the impacts (additive, synergistic, or antagonistic) of vestibular challenges and altered sleep/cognition on sensorimotor, behavioral, and performance outcomes, very few ground analog studies of human spaceflight studies have targeted multiple NASA Human Research Roadmap risks to characterize these interactions. This knowledge gap has limited the validation of countermeasures that can synergistically mitigate multiple risks. Additionally, following long-duration spaceflight, crewmembers often report an increase in fogginess. Yet the existing Spaceflight Standard Measures battery does not include outcome measures that can interpret the relative contributions of multiple hazards on sensorimotor declines. This proposal aims to characterize the interactive impacts of significant spaceflight hazards (i.e., cognitive overload, sleep deprivation, and altered gravity) on sensorimotor functions and associated adaptive or disruptive neural responses that explain these interactive effects. We will employ a within-subjects experimental design to evaluate the interactive effects of altered gravity and fatigue on sensorimotor standard measures. Participants will attend two separate and counterbalanced fatigue sessions: a 2-h cognitive fatigue session (Aim 1), and a total sleep deprivation (~28-h extended wakefulness) session (Aim 2). At various time points in each fatigue session, participants will undergo a series of sensorimotor tests under two altered gravity analog conditions: absence, and presence, of galvanic vestibular stimulation (GVS). The sensorimotor tests include (a) tandem walk, the obstacle walk, and the finger-nose motor-coordination task (based on NASA's Standard Measures for sensorimotor assessment and prior flight studies); (b) motor coordination and standardized balance tasks; and (c), two additional tasks (a dual-task tandem walk test and a hidden obstacle walk test, designed as transition walks between the sensorimotor tasks that participants will unknowingly perform). Across both aims, we will assess changes in performance, brain activation, and connectivity patterns, and workload perceptions using the NASA Task Load Index (TLX) questionnaire during the sensorimotor tasks. Study findings will facilitate (1) characterizing the interactions of fatigue and altered gravity on diverse types of sensorimotor functions (functional mobility vs. manual control); and (2) establishing new outcomes measures (i.e., variants of existing sensorimotor tests, workload perceptions) that may be included in the sensorimotor Spaceflight Standard Measures to elucidate these interactions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We will characterize the potential interactive impacts of multiple spaceflight hazards (fatigue, altered gravity), on sensorimotor, and operational outcomes and associated adaptive or disruptive neural responses that explain these interactive effects. We will establish additional outcome measures (e.g., workload perceptions; dual-task/DT tandem walk; hidden obstacle walk) that can assess the relative contributions of multiple stressors on sensorimotor impairments in astronauts.
Task Progress:	Building on the initial proposal, we have finalized our experimental setup to investigate the interactive effects of fatigue—induced by cognitive workload and sleep deprivation—and gravity on sensorimotor functions across 13 selected activities. Each activity has a distinct set of instructions and setup tailored to its specific demands. The selected activity set encompasses dynamic and static balance tasks, as well as eye-hand coordination activities. The five dynamic balance activities include: eyes open tandem walk, eyes closed tandem walk, dual task tandem walk, obstacle walk, and hidden obstacle walk. The tandem walk tasks are conducted on a straight path marked in the lab. For the obstacle walk, participants start from a sitting position, proceed along a straight path, turn around a cone, cross an adjustable obstacle (set to 90% of the participant's knee height), and return to the starting point. In the hidden obstacle walk, a unique arrangement is used where participants unknowingly encounter an obstacle as part of the experiment. The hidden obstacle walk path mirrors the normal obstacle walk path, with a 46mm-high obstacle that matches the floor's color to blend in. We have also organized six static balance tasks. In the first four, participants stand on a force plate surface with different conditions: eyes open, eyes closed, eyes open while performing a secondary task, and eyes closed while engaging in a secondary task. The final two static balance tasks involve participants standing on a 10cm foam surface (Dynamic Systems, Inc.), placed on the force plate, with eyes open in one task and closed in the other. Lastly, two activities focus on eye-hand coordination: the finger-to-nose coordination task and the Purdue Pegboard task. For the finger-to-nose coordination task, programmable buttons are used to record the number of clicks and timing, and each button is equipped with QR codes for further analysis using eye-tracking data. A preliminary session, lasting 90 minutes, is scheduled one week before the actual experimental session. During this session, participants provide informed consent, and their eligibility is assessed using the Pittsburgh Sleep Quality Index and the Fatigue Severity Scale, both administered via REDCap surveys. If deemed eligible, participants are given instructions on wearing the Empatica Embrace Plus smartwatch continuously until the experiment day. Participants are also briefed on completing a daily sleep diary, which is automatically sent and monitored via REDCap, with notifications to experimenters if sleep duration falls below a pre-set threshold. In addition, participants receive instructions on maintaining regular sleep patterns, avoiding excessive physical activity, and refraining from caffeine and alcohol in the 24 hours leading up to the experiment. The Empatica smartwatch tracks wearing time, pulse rate, skin temperature, electrodermal activity, accelerometer standard deviation, respiratory rate, and sleep status. These data are continuously uploaded to an online portal, allowing experimenters to closely monitor participants' activity in real time throughout the monitoring period. Finally, during the preliminary session, participants are trained on the 2-back test and all 13 sensorimotor activities. To streamline the process, preliminary and experimental sessions are conducted in parallel, with two lab spaces prepared for simultaneous use. Three experimenters have been trained to run the preliminary sessions on a rotating schedule. The experimental session takes place a week after the preliminary session and is scheduled for 7 hours for the cognitive fatigue group and 22 hours for the sleep deprivation group. In the cognitive fatigue session, the sensorimotor test battery is administered both before and after a two-hour fatigue induction period. For the sleep deprivation group, the test battery is repeated at five different time points: 1800 h, 2200 h, 0200 h (Day 2), 0600 h (Day 2), and 1000 h (Day 2). During each sensorimotor activity, participants perform four trials, with Galvanic Vestibular Stimulation (GVS) applied in two of those trials. The order of GVS and NoGVS trials is randomized, with half of the participants following a NoGVS-GVS sequence and the other half a GVS-NoGVS sequence. After each block of trials, participants' perception

	of workload is assessed using the NASA Task Load Index (TLX), and scores are recorded in REDCap for efficient data management. The bio-instrumentation setup for the experiment has been fully completed, ensuring that a wide range of physiological measures are recorded. Each measure has been carefully paired with its corresponding bio-instrument for optimal data collection. Body kinematics are being recorded using the Xsens full-body motion capture system (MVN, Xsens Technologies, Netherlands). Brain activity is captured with two 8x8 NIRSport2 functional near-infrared spectroscopy (fNIRS) systems (NIRx Medical Technologies, USA). Muscle activity is monitored via the Noraxon surface electromyography (EMG) system (Noraxon USA Inc.). Heart rate is tracked using the Actiheart-5 monitor (CamNtech Ltd, UK). Eye-tracking data is gathered using Tobii Glasses 3 (Tobii Technology, Sweden). Additionally, for static balance tasks, a multi-axis force plate system from AMTI (Advanced Mechanical Technology, Inc., USA) has been selected to record detailed force plate data. Data from the Xsens, fNIRS, EMG, and Tobii systems stream wirelessly into the main data collection PC, while the force plate transmits data via USB, and the Actiheart stores data internally. Each device was individually configured and tested against physical movements to verify the accuracy of the raw data. Once validated, all devices, except Actiheart, were set up to stream data into the Lab Streaming Layer (LSL) for synchronized data collection. LSL reduces the burden on the experimenter by allowing all data streams to be started and stopped with a single click. The accuracy of data from each device was tested with LSL individually before the final integration and re-tested after integrating all devices into LSL. We tested the accuracy of data synchronization after connecting all the devices and created a separate script to synchronize Actiheart data with other streams, using device timestamps. Additionally, we developed a monitoring code to oversee the quality of incoming data streams, ensuring any missing data is easily detected during the experiment. To further streamline the process, we automated the start and stop commands issued to the participant, syncing them with the data recording actions. For tasks 6 through 13, which are all 30-second tasks, the experimenter only needs to initiate the recording, as the computer will automatically stop the recording and issue the stop command to the participant after 30 seconds. After completing the experiment setup, we conducted four cognitive fatigue pilot sessions. Although the initial sessions took longer than expected, we successfully reduced the total duration to 7 hours by the third pilot. Some challenges encountered during these sessions included: participants reaching fatigue levels that caused them to refuse to continue, as well as dizziness induced by GVS. A data extraction pipeline was also created to streamline data from all bio instruments for the data analysis phase. With Institutional Review Board (IRB) approval secured, we have sent out a mass recruitment email to begin participant enrollment for the actual data collection. Currently, we are in the process of scheduling the first set of preliminary sessions.
Bibliography Type:	Description: (Last Updated: 05/27/2026)
Abstracts for Journals and Proceedings	Hayes J, Fairchild B, Nichols K, Jayasuriya J, Mehta RK. "Selective sensorimotor impairments with mental fatigue associated with G-transitions during and after spaceflight." ASPIRE - the HFES International Annual Meeting (Human Factors and Ergonomics Society), Phoenix, Arizona, September 9-13, 2024. Proceedings of the Human Factors and Ergonomics Society Annual Meeting, 0(0). https://doi.org/10.1177/10711813241275073 , Sep-2024